# Jessica U. Ezemba

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## EDUCATION

**Carnegie Mellon University** — Pittsburgh, PA
Ph.D., Mechanical Engineering | GPA: 3.92/4.0 — Expected May 2027
M.S., Integrated Innovation for Products & Services | GPA: 3.96/4.0 — Dec. 2022

**Texas A&M University** — College Station, TX
B.S., Mechanical Engineering | GPA: 3.52/4.0 — May 2021

## SKILLS

**Programming & Engineering:** Python, FEniCSx, PyAnsys, ANSYS, Siemens NX, Simcenter 3D, MATLAB, SOLIDWORKS, CATIA, AutoCAD, PTC Creo
**AI/ML:** PyTorch, PyTorch Geometric, Multi-Agent Systems, LangGraph, Vision Transformers, GNNs, Geometric Deep Learning, Autoencoders, LLMs, RAG, MCP
**Computation & DevOps:** HPC, FEA, CFD, Probabilistic Simulation, Surrogate Modeling, Neural Operators

## RESEARCH EXPERIENCE

**Carnegie Mellon University** — Pittsburgh, PA
*PhD Candidate, Mechanical Engineering* — Aug. 2023 - Present

- Built custom Multi-Agent System (MAS) for engineering design and simulation utilizing organizational design principles, increasing computational efficiency by 50%
- Developed Model Context Protocol (MCP) server for prompt-to-CAD, parametric geometry variation, and FEA/CFD simulations, enabling faster integration into MAS and reducing computational costs by 50%
- Created comprehensive benchmark datasets for evaluating LLM performance on engineering simulation reasoning across text, image, and video modalities (200,000+ questions, 10,000 simulations)
- Engineered neural network surrogate models with FEniCSx to quantify uncertainty in complex engineering simulations, reducing computational costs by 90%
- Applied Geometric Deep Learning with custom GNN data loaders for complex 3D unstructured meshes

## PROFESSIONAL EXPERIENCE

**Siemens** — Remote / Princeton, NJ
*PhD Research Intern, Pre-Research and Development* — Jan. 2026 - Present

- Built an end-to-end CAD/CAE pipeline connecting AI agents to Siemens NX and Simcenter 3D via a 34-tool MCP server, optimizing designs against yielding, fatigue, buckling, and vibration requirements
- Designed ontology-guided engineering AI agents that modify CAD geometry through feature-based edits (ribs, fillets, pockets), achieving 2.5x the mass reduction of an unstructured baseline agent (9.3% vs. 3.7%)
- Benchmarked 17 AI models from 6 vendors on 550 structural FEA questions; none exceeded 67% accuracy

**KLA** — Ann Arbor, MI
*Advanced Mechatronics Intern* — May 2022 - Aug. 2022 & May 2023 - Aug. 2023

- Developed a 3D parametric geometry and physics-based optimizer to reduce the time to design flexures
- Created a 3D design exploration algorithm using ANSYS for fatigue, buckling, and frequency analysis
- Developed an optimization algorithm in MATLAB to decrease controller tuning time by over 50%

**Symbotic** — Wilmington, MA
*Systems Design Intern* — Jan. 2023 - May 2023

- Increased warehouse automation throughput by using FEA to optimize component mass and structural integrity
- Designed high-risk safety devices for warehouse staff, achieving 100% compliance with safety regulations

## SELECTED PUBLICATIONS

1. **Ezemba, J.,** et al. (2026). "Organizationally-Intelligent Agents: Connecting Multi-Agent Engineering Systems to Organizational Design Theory." Design Computing and Cognition '26, Springer Nature.
2. **Ezemba, J.,** McComb, C., & Tucker, C. (2026). "Structure Shapes Coordination: An Empirical Evaluation of Organizational Design Principles in Multi-Agent Engineering Design." ASME IDETC 2026.
3. Mehta, M.D.*, **Ezemba, J.*,** et al. (2026). "VizArQA: A Benchmark for Visual Question Answering in Architectural Simulation." J. of Eng. for Sustainable Buildings and Cities.
4. **Ezemba, J.,** Pohl, J., Tucker, C., & McComb, C. (2026). "A Large-Scale Benchmark to Assess Vision-Language Model QA Capabilities in Engineering Simulations." Communications Engineering (Nature Portfolio).
5. **Ezemba, J.,** McComb, C., & Tucker, C. (2026). "Neural Network Surrogate Modeling for SFEM Using 3D Graph Representations." J. of Mechanical Design, Vol. 148, No. 1.